Class Imbalance in Histopathology: Prevalence Alone Costs Several Points on BRACS

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Abstract

In a long-tailed training set, the rare classes usually receive both a smaller share of the patches and fewer patients. This report isolates the first factor, class prevalence, at fixed patient support on BRACS, a seven-class breast-lesion task, and repeats the design of the TCGA-UT study of this series. A multinomial logistic classifier on frozen Virchow2 features is trained on the same ten patients per lesion class and the same total of 2,240 patches in every arm; only the share of the budget that each class receives changes, following an exponential long-tail profile with imbalance ratio ρ from 1 to 100 and a random class order per draw. A further arm uses the classes' native training shares, which realize $\rho = 6.7$. On BRACS, prevalence damages balanced accuracy early and steadily: the loss is 1.15 percentage points already at $\rho = 2$ (95% interval: 0.14 to 2.35), which meets the one-point threshold, 3.86 points at $\rho = 10$ (2.33 to 5.44), and 7.98 points at $\rho = 100$ (6.01 to 10.17), from a balanced level of 39.57 percent. The loss per doubling of ρ is about 1.2 points over the whole grid and does not accelerate. At $\rho = 100$ the two tail classes fall from 38.7 to 6.0 percent recall. The native allocation costs 4.02 points (2.68 to 5.52), as much as the patient-count gap between five and twenty patients on this dataset, and lies on the ratio curve. Raw probability quality degrades sharply, with macro negative log-likelihood rising from 1.56 to 4.00 nats at $\rho = 100$, but a single validation temperature removes 91 percent of that increase and 88 percent of the increase in expected calibration error. On BRACS, prevalence at fixed patient support is therefore a material source of discrimination loss from the mildest imbalance on, whereas its calibration damage is again mostly a matter of scale.

1 Motivation

The benchmark of this series found that severe class imbalance damages balanced accuracy on BRACS by 4.30 to 13.23 percentage points, depending on which classes form the tail, and that validation temperature scaling removes most of the accompanying differences in probability quality (Queisler, 2026b,a). As on TCGA-UT, these numbers bundle several factors: each imbalanced condition changed class prevalence and the patients behind each class at the same time, only three allocations were measured, and the classifier was a two-layer perceptron. The TCGA-UT study of this series separated prevalence from those factors and found a slow, accelerating loss that reaches one point only at $\rho \approx 20$ and stays below three points at $\rho = 100$ (Queisler, 2026e). That result does not transfer automatically. BRACS has seven lesion classes instead of 30 cancer types, several of them lie on a morphological continuum from benign to malignant, and balanced accuracy on the same features is about 40 instead of 72 percent (Queisler, 2026c). Heavily overlapping classes make the decision boundaries more sensitive to the class prior.

This report therefore asks the TCGA-UT question on BRACS: *how fast do balanced accuracy, negative log-likelihood, and calibration error degrade as the training class distribution moves from balanced to long-tailed, when the patients of every class and the total patch budget stay fixed, and where does the dataset's native distribution lie on that curve?* The design, endpoints, and prestated readings follow the TCGA-UT study, with one deviation in the number of patients per

class that the BRACS patient pool forces. Patient shortage and patch shortage per class remain outside the design.

2 Design

The study uses the seven BRACS lesion subtypes, from normal tissue (N) through pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), and ductal carcinoma in situ (DCIS) to invasive carcinoma (IC). Features are frozen 2,560-dimensional Virchow2 embeddings of 256×256 patches (Zimmermann et al., 2024). The three fixed patient-disjoint splits of the benchmark each hold about 103 training, 24 validation, and 24 test patients, and each split is drawn ten times. Validation and test partitions keep their natural class distribution. In each draw, G patients per class are sampled from those with at least $D = 160$ training patches, and the same patients enter every arm of the draw. Each patient's patches follow one fixed round-robin order over slides, and an arm that assigns m patches to a patient takes the first m of them, so a class's smaller allocation is always contained in its larger one.

The TCGA-UT study used $G = 20$. On BRACS, only 10 to 13 ADH patients per training split reach the depth of 160 patches, so this study uses $G = 10$ for every class and keeps the depth, which the most extreme head allocation needs. Every arm trains on the same total budget $T = K \cdot G \cdot 32 = 2{,}240$ patches for $K = 7$ classes, which is 32 patches per patient in the balanced arm, as on TCGA-UT. A ratio arm with imbalance ratio ρ assigns the class at rank $k \in \{0, \ldots, K-1\}$ the count

$$N_k = N_0 \, \rho^{-k/(K-1)}, \tag{1}$$

with N_0 chosen so that the counts sum to T, rounded to integers that keep the sum, at least one patch per patient, and at most D per patient. This exponential profile is the allocator of the benchmark (Queisler, 2026a). It moves the class prior and the tail support together: at $\rho = 100$ the head class receives 1,206 patches, about 121 per patient, and the tail class 12, between one and two per patient. With only seven classes, neighbouring ranks differ by a factor of $\rho^{1/6}$, which is 2.15 at $\rho = 100$, against 1.17 for the 30 classes of TCGA-UT. The assignment of classes to ranks is a random permutation drawn per split and draw, so each lesion class visits head, body, and tail across the 30 draws. Within a class, the count is spread as evenly as possible over its ten patients.

The native arm N gives each class its share of all training patches of the split, scaled to T, in the classes' own order and with the same ten patients. IC, PB, and DCIS receive between 431 and 750 patches, whereas N, ADH, FEA, and UDH receive between 90 and 188, and the realized ratio averages 6.7 over the 30 draws. Table 1 lists the arms, and Figure 1 shows every realized class distribution.

Each arm fits a multinomial logistic classifier whose regularization strength is selected from nine values between 10^{-6} and 100 by validation patient-macro balanced accuracy (Queisler, 2026d). After selection, one temperature is fitted on the validation logits by minimizing their negative log-likelihood and applied to the test logits (Guo et al., 2017); it rescales confidence and leaves predictions unchanged. The study fits eight arms for each of 30 split-draw pairs, 240 classifiers in total, and every fit converged. Test data never entered selection or temperature fitting.

3 Evaluation

The endpoints are those of the TCGA-UT study (Queisler, 2026e). The primary endpoint is the *damage* $\Delta(\rho) = \mathrm{BA}(1) - \mathrm{BA}(\rho)$, where BA is patient-macro balanced accuracy on the test partition, computed as the mean of per-class patch recalls in which every test patient of a class carries equal weight. Estimates are pooled over splits and draws with equal weight per split.

Arm	Realized ρ	Head class	Tail class	Tail per patient
r1	1.0	320	320	32
r2	2.0	441	220	22.0
r5	5.0	622	124	12.4
r10	9.9	766	77	7.7
r20	20.2	908	45	4.5
r50	49.3	1,084	22	2.2
r100	100.5	1,206	12	1.2
N	6.7	625–750	90–115	9.0–11.5

Table 1: All arms train on the same ten patients per class and the same 2,240 patches; only the per-class share changes. Head and tail are the largest and smallest class counts in patches. The realized ratio is averaged over 30 split-draw pairs and departs from the nominal ratio only through integer rounding. The native arm N uses the training split's own class shares, so its extreme counts differ between the three splits; the ranges span them.

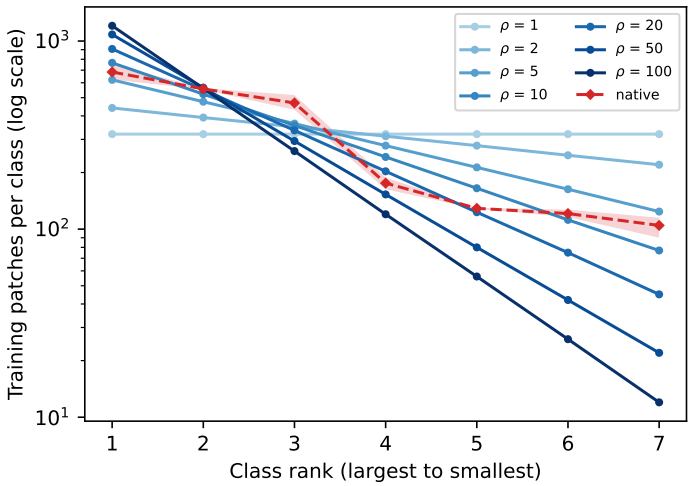


Figure 1: The ratio arms tilt one fixed budget around the second and third classes, whereas the native allocation splits into three large and four small classes. Training patches per class, sorted from the largest to the smallest class, on a logarithmic axis, so an exponential profile is a straight line. Blue lines are the ratio arms, from light ($\rho = 1$) to dark ($\rho = 100$); their sorted counts are identical in every draw. The red dashed line is the native arm, averaged over the 30 split-draw pairs, with the shaded band spanning the smallest and largest count at each rank. All arms sum to 2,240 patches.

Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals. Prevalence counts as damaging at ratio ρ if $\Delta(\rho)$ is at least one percentage point and its interval lies entirely above zero; the *onset* is the smallest ratio of the grid that meets this criterion. The *slope* is the least-squares slope of BA on $\log_2 \rho$ per replicate, in points per doubling of ρ, over the whole grid and separately over $\rho \leq 10$ and $\rho \geq 10$.

Probability quality is measured by the macro negative log-likelihood (NLL), which averages the log-loss of the true class with equal weight for every test patient and class, and by the expected calibration error (ECE) of the top-label confidence in ten fixed bins with patient weights, both with and without temperature scaling. Both depend on the class distribution of the evaluation data (Mosquera et al., 2024), which is the same for all arms. With about 24 test patients per split, the ECE of BRACS is estimated much less precisely than on TCGA-UT. Two diagnostics complete the evaluation. The recall of the head, body, and tail classes by assigned rank shows where the damage lands; with seven classes, these groups hold three, two, and two classes. The native arm is compared with the balanced arm and with the ratio curve, interpolated linearly in ρ at N's realized ratio.

The three readings fixed in advance are those of the TCGA-UT study. If Δ stays below one point up to $\rho = 100$, prevalence alone is not a material deficit on BRACS at fixed patient support. If Δ crosses the threshold within the grid, the onset and slope locate the ratio from which a prevalence correction is worthwhile. If the native arm lies below the curve, the native order of the classes, which ties prevalence to specific lesion types, costs more than a random order.

4 Results

The results follow the endpoints: first the accuracy curve with its onset and slope, then the classes on which the damage lands, the position of the native allocation, and finally probability quality before and after temperature scaling. The balanced arm reaches 39.57 percent balanced accuracy (36.70 to 42.53). Ten real patients with sixteen patches each reached 38.49 percent in Queisler (2026c); twice the patch depth of this design adds about one point to the balanced reference, a plausible size for a cross-study comparison with different draws.

4.1 Accuracy curve

Balanced accuracy falls monotonically with the imbalance ratio, and it falls from the first step on (Table 2, Figure 2). At $\rho = 2$ the damage is 1.15 points (0.14 to 2.35), so the onset of the grid is its smallest ratio, although the lower bound of the interval lies well below one point. The damage reaches 2.97 points (1.62 to 4.46) at $\rho = 5$, 3.86 points (2.33 to 5.44) at $\rho = 10$, and 7.98 points (6.01 to 10.17) at $\rho = 100$, where the classifier has lost a fifth of its balanced accuracy. From $\rho = 5$ on, every interval lies above the threshold. The three splits agree in direction and size: the damage at $\rho = 100$ is 6.98, 8.67, and 8.28 points.

The loss per doubling of ρ is 1.18 points (0.90 to 1.50) over the whole grid, three times the 0.39 points of TCGA-UT (Queisler, 2026e). Unlike on TCGA-UT, the curve is close to linear in $\log \rho$: up to $\rho = 10$ it loses 1.19 points per doubling (0.74 to 1.65), from $\rho = 10$ to $\rho = 100$ it loses 1.23 points (0.89 to 1.64), and the two intervals overlap widely. About half of the damage at $\rho = 100$ is already present at $\rho = 10$.

4.2 Where the damage lands

The aggregate damage again results from recall moving from the rare to the frequent classes, but the transfer is far larger than on TCGA-UT (Table 3). In the balanced arm, the three head

Arm	BA (%)	Damage Δ	Macro NLL (nats) raw	Macro NLL (nats) scaled	ECE (points) raw	ECE (points) scaled	Temperature
r1	39.57	—	1.565	1.576	12.29	6.60	0.65
r2	38.42	**1.15** [0.14, 2.35]	1.581	1.599	8.31	6.59	0.86
r5	36.60	**2.97** [1.62, 4.46]	1.878	1.642	10.94	6.35	1.54
r10	35.71	**3.86** [2.33, 5.44]	1.938	1.668	13.15	6.41	1.67
r20	34.49	**5.08** [3.34, 6.94]	2.509	1.700	19.55	6.87	2.45
r50	32.97	**6.60** [4.76, 8.65]	3.218	1.752	25.11	8.00	3.54
r100	31.59	**7.98** [6.01, 10.17]	4.004	1.796	30.09	8.77	4.43
N ($\rho = 6.7$)	35.55	**4.02** [2.68, 5.52]	1.984	1.692	8.82	5.38	1.53

Table 2: On BRACS, accuracy degrades from the mildest imbalance on, raw probability quality collapses, and temperature scaling removes almost all of the probability-quality loss. Balanced accuracy (BA) is test patient-macro balanced accuracy; the damage is $\mathrm{BA}(1) - \mathrm{BA}(\rho)$ in percentage points with paired 95% bootstrap intervals, in bold where it meets the one-point criterion. NLL and ECE are shown without (raw) and with (scaled) the validation temperature; the temperature column is the mean fitted temperature over 30 fits, where values above one soften overconfident predictions and values below one sharpen underconfident ones. The 95% intervals of the absolute BA levels are 5 to 6 points wide and are shared by all arms.

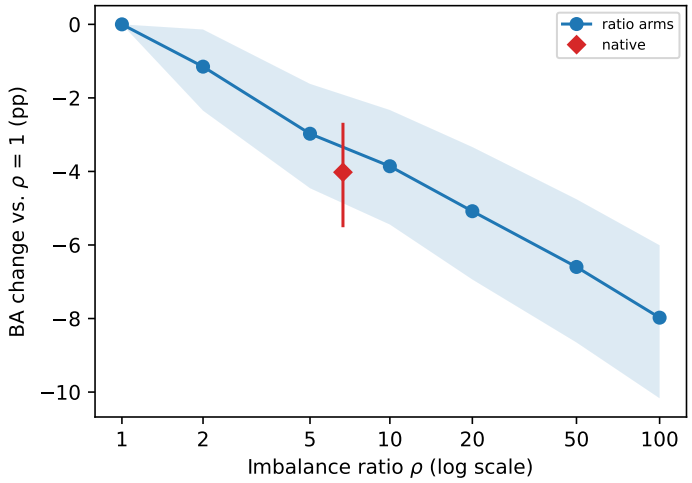


Figure 2: Balanced accuracy on BRACS degrades steadily, by about one point per doubling of the imbalance ratio. The line shows the change in patient-macro balanced accuracy relative to the balanced arm, with the ratio arms at their realized ratio and a shaded paired 95% bootstrap band; the red diamond marks the native allocation at its realized ratio of 6.7.

classes, two body classes, and two tail classes of the random rank order reach between 37 and 44 percent recall; with only seven classes and 30 draws, the rank groups do not average out class difficulty as completely as on TCGA-UT. With growing ρ, the head classes gain up to 15.0 points, whereas the body classes lose 17.7 points and the tail classes 32.7 points. At $\rho = 100$ the tail classes reach 6.0 percent recall, so the classifier almost never predicts them. The body classes lose as well because, with seven classes, the exponential profile gives them only 120 and 56 of the 320 patches of the balanced share at $\rho = 100$. Much of the redistribution happens early: at $\rho = 2$ the tail classes already lose 12.7 points and the head classes gain 8.5.

Rank group	r1	r2	r5	r10	r20	r50	r100
Head (3 classes)	37.41	45.89	48.07	50.01	50.36	51.70	52.36
Body (2 classes)	43.66	39.65	36.75	34.48	32.94	29.37	26.00
Tail (2 classes)	38.73	25.98	19.23	15.50	12.23	8.49	6.04

Table 3: Imbalance moves recall from the rare to the frequent BRACS classes, and the tail classes are almost never predicted at $\rho = 100$. Mean patient-macro recall in percent of the head, body, and tail classes of each draw, by their assigned rank, averaged over 30 fits. The native arm is omitted because its class order is not the random rank order that defines the groups.

4.3 Native allocation

The native class distribution of BRACS costs 4.02 points against the balanced arm (2.68 to 5.52), four times the threshold, although its realized ratio of 6.7 is modest. It lies on the ratio curve: interpolated at its realized ratio, the curve predicts a damage of 3.27 points, and the native arm falls 0.75 points below it (-1.99 to 0.47), a difference whose interval contains zero. The native profile has two levels rather than an exponential decay, with three frequent classes (IC, PB, DCIS) and four rare ones (N, ADH, FEA, UDH), and it assigns the rare share to specific lesion types, among them the atypical lesions ADH and FEA. Within the precision of this study, the realized ratio alone nevertheless places it on the curve, and the study finds no evidence that the native order of the classes costs more than a random order.

4.4 Probability quality

Raw probability quality degrades far more than accuracy, and a single temperature again removes almost all of the degradation (Table 2, Figure 3). Macro NLL rises from 1.56 nats in the balanced arm to 4.00 nats at $\rho = 100$, an increase of 2.44 nats (1.73 to 3.26). The raw ECE is not monotone in ρ. The balanced arm is underconfident, with a mean temperature of 0.65 and a raw ECE of 12.29 points; mild imbalance lowers the ECE at $\rho = 2$ by 3.98 points (0.33 to 6.74). From $\rho = 5$ on the mean temperature exceeds one, so overconfidence takes over; the ECE returns above the balanced level at $\rho = 10$ and rises to 30.09 points at $\rho = 100$, an increase of 17.81 points (8.91 to 26.79) over the balanced arm. After temperature scaling, the increase at $\rho = 100$ shrinks to 0.22 nats of NLL (0.17 to 0.27) and 2.17 points of ECE (-1.94 to 5.92), whose interval contains zero. Scaling therefore removes 91 percent of the NLL increase and 88 percent of the ECE increase. The remaining scaled NLL increase is resolved from $\rho = 2$ on and follows the accuracy loss. In the balanced arm, scaling raises the test NLL slightly, from 1.565 to 1.576 nats, because the temperature is fitted on a validation partition of only about 24 patients. The native arm follows the same pattern: its raw NLL exceeds that of the balanced arm by 0.42 nats (0.28 to 0.60) and its scaled NLL by 0.12 nats (0.08 to 0.16), while its ECE differs from the balanced arm by -3.47 points raw (-9.55 to 4.02) and -1.22 points scaled (-4.59 to 1.03).

The fitted temperatures and the selected regularization show the source of the overconfidence. The temperature rises from 0.65 in the balanced arm to 4.43 at $\rho = 100$, so predictions must

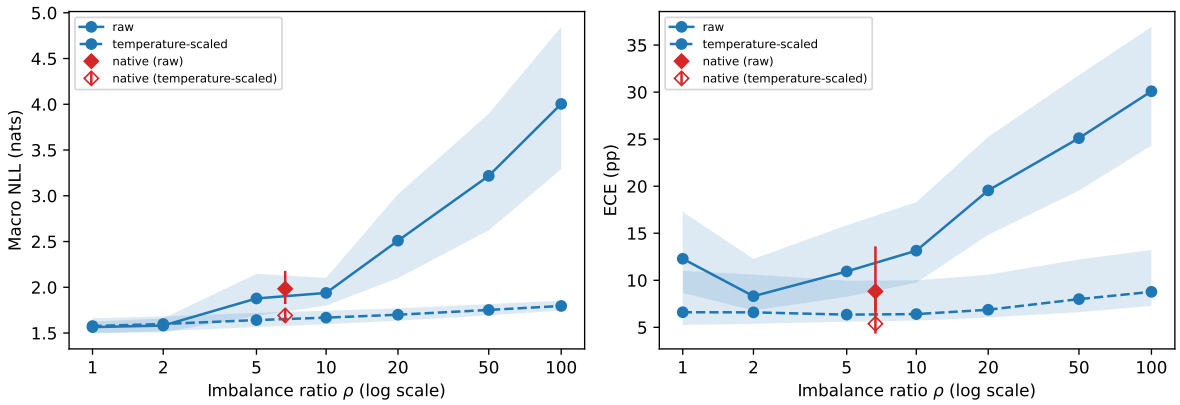


Figure 3: Raw probability quality on BRACS collapses with imbalance but recovers almost completely under temperature scaling. Left: macro NLL; right: ECE; both raw (solid, filled) and after validation temperature scaling (dashed, open). Blue lines connect the ratio arms at their realized ratio, with shaded paired 95% bootstrap bands; the red diamonds mark the native allocation at its realized ratio of 6.7.

be softened by more than a factor of four. The balanced arm selects $\lambda = 1$ or 10 in all 30 fits, whereas $\rho = 100$ selects $\lambda \leq 10^{-3}$ in 11 of 30 fits and the smallest grid value, 10^{-6}, in 4; at $\rho = 50$ and $\rho = 20$ the smallest value is selected in 3 and 1 fits. When these eight fits are refitted with λ down to 10^{-10}, validation selects a smaller value in seven of them, but selection on the extended grid changes the damage by at most 0.05 points at any ratio. Much of the validation gain below 10^{-6} reflects the solver tolerance rather than the regularization: refitted to a gradient tolerance of 10^{-10}, the validation curves flatten below 10^{-6}, and selection on these fits changes the damage by less than 0.01 points. The grid boundary therefore does not affect the results. Weak regularization lets the logits grow, which produces overconfident errors on the rarely predicted tail classes and explains the large raw NLL. As on TCGA-UT, the study does not separate this effect of model selection from the direct effect of the skewed class prior on confidence.

5 Discussion

Of the three readings fixed in Section 3, the results select the second, in a much stronger form than on TCGA-UT: prevalence alone damages discrimination on BRACS from $\rho = 2$ on, by about one point per doubling, and by 7.98 points at $\rho = 100$. The third reading is not supported, because the native allocation lies on the curve. The contrast with TCGA-UT is large. At $\rho = 10$, the tail classes of both datasets keep about eight patches per patient, yet the damage is 4.02 points on BRACS against 0.57 on TCGA-UT (Queisler, 2026e); at $\rho = 2$, where the tail keeps 22 patches per patient, BRACS already loses more than TCGA-UT does at $\rho = 20$. The loss of per-patient support in the tail therefore cannot account for the BRACS damage by itself. A more likely source is the overlap of the BRACS classes. When classes overlap as much as a balanced accuracy of 40 percent implies, a skewed training prior shifts the decision boundaries between neighbouring lesion types far into the rare classes, whereas well-separated cancer types keep their boundaries almost regardless of the prior. The steeper profile of seven classes, with a factor of 2.15 between neighbouring ranks at $\rho = 100$, adds to this. The study does not test the overlap explanation directly.

For the prevalence factor on BRACS, the curve refines the benchmark's three-allocation readout. The benchmark measured 4.30 to 13.23 points of concentrated severe-imbalance damage across tail assignments (Queisler, 2026b); the 7.98 points at $\rho = 100$, averaged over random class

orders, lie inside that range. Because the benchmark also changed the patient pool, budget, and classifier, the magnitudes are not directly comparable, but both place severe prevalence damage on BRACS at several points. The native distribution costs 4.02 points, which is of the same size as the 4.80-point gain from five to twenty patients per class on BRACS (Queisler, 2026c). On TCGA-UT, the native prevalence cost was about an eighth of the corresponding patient-count gain. On BRACS, class prevalence and patient count are therefore deficits of comparable size, and a long-tailed BRACS training set would need a prevalence correction even at adequate patient support.

Calibration behaves as on TCGA-UT, only more extremely. The raw NLL rises about 2.6-fold at $\rho = 100$ and the ECE about 2.4-fold, while one validation temperature removes almost all of both increases. The calibration damage of imbalance on BRACS is thus again mostly a uniform overconfidence, amplified by weakly regularized models that validation accuracy selects under strong imbalance, and not a class-specific distortion. Prevalence-based corrections such as class weighting or threshold moving (Buda et al., 2018) are more promising for discrimination on BRACS than on TCGA-UT, because much of the lost tail recall appears to be a boundary shift that a prior correction targets; for probability quality, they would again have to beat a temperature-scaled baseline.

With this study, the prevalence factor is measured on both datasets of the series. The remaining split studies vary patch count per class at fixed prevalence and patient count at fixed patch count, one factor at a time and with the same plain baseline, to show which of the benchmark's factors carries the damage it measured.

6 Limitations

Each ratio moves the class prior and the tail support together, because the total budget is fixed; the study does not separate the effect of prevalence as such from the smaller per-patient patch count of the tail classes. The comparison with TCGA-UT at matched tail support argues that the prior carries most of the BRACS damage, but a design at fixed tail support would test this directly. The study uses ten patients per class instead of twenty, because only 10 to 13 ADH patients per training split reach 160 patches; the ADH patients therefore hardly change between draws, and the draws are less independent for this class than for the others. The curve describes imbalance at this fixed patient support and says nothing about the combination of rare classes with few patients.

The test and validation partitions hold only about 24 patients per split, which makes the ECE estimates wide and the temperature fits noisy; the non-monotone raw ECE at small ratios is resolved only at $\rho = 2$. The rank groups contain two or three classes, so their recall mixes the imbalance effect with the difficulty of the particular classes that drew each rank. The native arm is a single allocation, and its agreement with the curve does not show that every class order behaves like a random one. All results are conditional on frozen Virchow2 features, multinomial logistic regression, BRACS, and three overlapping splits.

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